Digital Product Passport as Digital Carrier for Information of Life Cycle Assessment: A Feasibility Study of Solvolysis on Composite Recycling for Wind Turbines Blades

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Abstract. The study incorporates the concept of a digital product passport as a data carrier for tracking the end-of-life of wind turbine blades across the processes value chain, integrating circular economy principles into the business model. Evaluation of a novel chemical recycling process for wind turbine blades using Life Cycle Assessment (LCA) and Life Cycle Cost Analysis (LCCA) suggests that the chemical recycling process has significantly reduces environmental impacts when compared to landfill disposal and offers potential economic benefits. However, scalability and material recovery efficiency remain key factors for widespread adoption.

Keywords: Wind turbine blades · Life Cycle Assessment · Life Cycle Cost Analysis · Digital Product Passport · Sustainability · Circular Economy · Eco-design

1 Introduction

As wind energy demand increases, the end-of-life management of wind turbine blades remains a challenge due to their composite structure. The shift to renewable energy sources is crucial for managing climate change. Wind energy, in particular, plays an important role and the blades of the wind turbines are key components. Blades contain a complex material mix that includes glass fiber, thermosets like epoxy, thermoplastics like PVC, PET and PU, balsa wood, adhesives, and coatings. Due to this complex mix, blades pose a challenge for material recycling since it's difficult to separate them [1]. Epoxy resin, favored for its strength and lightweight properties, enhances the durability and structural integrity of these blades. However, the difficulty in separating these materials complicates recycling efforts.

Currently, end-of-life wind turbines can be recovered up to 80% on average through standard recycling routes [2]. Although the blades and composite materials remain the main challenge for the wind sector, they can be recycled through thermal, chemical, or mechanical processes. Thermal recycling includes heating the blades at high temperatures, mechanical recycling reduces blades to particles, and chemical recycling employs

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solvents to break down the components for reuse. Despite efficiency and research regarding the chemical process, industrial-scale implementation remains limited due to high energy requirements and high-cost catalysts for the chemical process. A limited number of, recent studies show that recovering bisphenol A (BPA) from epoxy resins can boost recycling efficiency while minimizing energy consumption. [3] leading to increased value for the end product.

This preliminary theoretical study investigates the economic and environmental impacts of solvolysis as a method for separating epoxy from fibers in wind turbine blade recycling on an industrial scale. Upcoming EU legislation is expected to force companies to disclose information related to recycled materials and the recyclability ratio of products through the Digital Product Passport (DPP) [4]. Therefore, blockchain technology could improve the DPP by incorporating raw material data, in a secure, trusted and decentralized way, supporting further the recycling efforts from digital perspective [5]. Conclusively, the research aims to explore the feasibility and development of a business model framework prioritizing material recovery, recycling, and sustainability, while focusing on waste hierarchy and reusing activities.

2 Literature Review

2.1 Wind Turbine Blade Waste Management

The wind industry is a sustainable, cost-effective and clean energy source. EU's commitment to reducing greenhouse gas emissions by up to 95% in the coming decades, imposes that wind turbines are likely to play an important part in the future energy supply [6].

The life expectancy of wind turbines being around 25 years, along with the forecasted high demand for wind turbines, estimates that 325 kilotons of wind turbine blades-of-life in EU the forthcoming decade. For that reason, there is ongoing research based on waste management prioritizing circularity and more specifically on blades due to their complexity. Most wind turbines contain steel, copper and aluminum meaning that almost 94% can be recycled [7].

The disposal of decommissioned turbine blades involves significant environmental problems, such as the generation of microplastics and pollution of soil and water sources and economic challenges for financing the operations for the decommission [6]. Traditional waste management methods, such as landfilling and incineration, are considered unsustainable due to their harmful effect on the environment and public health. As a result, there is an urgent need for creative and sustainable waste management techniques that are consistent with the concepts of a circular economy, which prioritizes resource recovery, recycling, and waste reduction [8].

Current research identifies many approaches that handle end-of-Life wind turbine blades, including mechanical recycling, thermal processing, and repurposing for new uses [6]. However, technological, economic, and legal barriers frequently impede the application of these approaches. To successfully address the rising waste issue, it's essential to develop composite materials that are recyclable at the end of their life cycle [7].

2.2 Recycling

Effective recycling solutions are required to reduce environmental impact, recover valuable resources, and promote sustainability in the sector [9]. To solve these issues, a variety of recycling techniques have been developed, each with its own set of advantages and limits.

Mechanical

Mechanical recycling is one of the most popular ways of treating end-of-life wind turbine blades. Typically, mechanical shredders are used to break down composite materials into smaller particles. One of the key benefits of mechanical recycling is that it uses less energy than other recycling technologies, making it an economically viable choice. However, mechanical recycling has its limits. The quality and consistency of the recovered materials might vary because the procedure might not sufficiently separate the various components of the composite materials. Furthermore, recycled materials may have lower mechanical qualities than virgin materials, limiting their usage in vital uses [10].

Chemical

Chemical recycling known also as solvolysis is a cutting-edge way of recycling end-of-life wind turbine blades, using chemical processes to break down composite materials into each of their components. This method not only improves the quality of the recovered components, but it also allows for the regeneration of the resin, which can then be reused in new composite compositions. However, chemical recycling can be more complicated and energy-intensive than mechanical or thermal techniques. The need for specialized equipment and well-defined procedures might result in greater operational expenses, which might limit adoption. Furthermore, the quality and consistency of recovered materials are strongly reliant on the exact chemical procedures used, demanding stringent quality control techniques [10].

Thermal

Thermal recycling refers to a variety of methods, including pyrolysis and oxidation. This recycling process comes along with increased energy consumption and operating expenses. The amount of energy required to heat the materials can be large, raising questions about the process's overall sustainability. Furthermore, the possible losses during heat degradation might restrict the recovery of useful materials [10].

An alternative to pyrolysis is co-processing. Co-processing is a waste management technique that uses combustion to convert waste products into fuel and raw material. This process promotes a circular economy since it helps to conserve energy and resources in other material production processes, reducing industry dependency on raw and virgin materials that are consequently decreasing at an unsustainable rate [9].

In conclusion, the effectiveness of wind turbine blade recycling methods varies based on their impact over time. Mechanical recycling is a viable short-term option because of its low energy consumption and economic feasibility, but the low grade of recovered materials restricts its long-term sustainability. Thermal recycling, which includes pyrolysis and oxidation, is a medium-term solution that balances material recovery with increased energy needs and operating costs. However, worries about sustainability and efficiency might prevent broad implementation. Chemical recycling, despite its complexity and energy-intensive nature, provides a long-term solution by recovering high-quality

materials which makes it a potential technique for a circular economy. As the wind energy sector strives to increase sustainability, advances in recycling technology and regulatory guidance will be critical in optimizing these processes for broad adoption.

2.3 Legislation Framework for Wind Turbine Blades

The legislative framework governing wind turbine blade waste management in the European Union is primarily shaped by the European Waste Framework Directive (2008/98/EC). The framework establishes a waste hierarchy that prioritizes prevention, reuse, recycling, recovery, and disposal. This directive serves as the basis for national waste policy, pushing the wind sector to adopt circular economic concepts. Despite the directive's broad scope, it lacks precise recycling objectives for composite components, like wind turbine blades. In response to increasing decommissioning volumes, different member states are implementing specific measures. For example, France is proposing regulating measures that would mandate wind turbine recycling, which might create an example for other countries [11]. The Circular economy action plan [12] strengthens this transition by encouraging the recycling of complex materials and discourages landfill disposal, indicating a steady regulatory move toward more strict sustainability criteria.

A key policy instrument that could enhance wind turbine blade waste management is Extended Producer Responsibility (EPR), which shifts waste management responsibilities from governments to manufacturers. EPR has been widely applied in industries such as electronics and packaging, resulting in increased recycling rates and lower waste disposal costs [13]. However, its application to wind turbines is hampered by regulatory exemptions. Wind turbine blades are designated as Large-Scale Fixed Installations (LSFIs), so they are excluded from the Waste Electrical and Electronic Equipment (WEEE) Directive. [11] legal gap restricts industry-wide responsibility for blade end-of-life management, leaving disposal decisions primarily up to project owners rather than manufacturers. Furthermore, the lack of a developed market for secondary materials generated by decommissioned blades, along with high recycling prices, creates financial and logistical obstacles.

Emerging regulatory frameworks, such as the Digital Product Passport (DPP), present opportunities to improve traceability and transparency in wind turbine blade waste management. The DPP is intended to give extensive product life cycle information, such as material composition, repairability, and disposal alternatives [4]. By including standardized "track and trace" data, the DPP connects with programs like the Sustainable Products Initiative (SPI) and the Eco-design Directive, which aim to set minimum sustainability criteria across industries [14]. Nevertheless, obstacles remain in integrating the DPP into current legal frameworks, especially in terms of identifying data ownership, maintaining accessibility between EU member states, and aligning with business demands. If effectively implemented, the DPP might lead to more efficient recycling, material recovery, and regulatory compliance, complementing the EU's overall goal of a circular economy [5]. The advancement of these legal frameworks will help shape the future of wind turbine blade waste management, bridge present policy gaps, and ensure long-term end-of-life solutions for renewable energy infrastructure.

3 Methods

3.1 Life Cycle Assessment

Life Cycle Assessment (LCA) is a systematic process for assessing the environmental implications of a product at all phases of its life, from raw material extraction to manufacture, use, and disposal. The purpose of LCA is to offer a complete perspective of a product or service's environmental elements and prospective consequences, assisting in the identification of chances for improvement and supporting decision-making processes.

Furthermore, LCA is regulated by international standards like ISO 14040 and ISO 14044, which give a structure and procedures for performing LCA research. The Life Cycle Analysis (LCA) process begins with Goal and Scope Definition, defining the study's objectives and analysis boundaries. The Life Cycle Inventory (LCI) collects data on energy, material inputs, and environmental discharges. Then, the Life Cycle Impact Assessment (LCIA) assesses environmental implications, including global warming potential and resource depletion. Lastly, the Interpretation phase analyzes the data to make informed judgments, detect key concerns, and provide suggestions [15].

The functional unit for this study is defined as (1) wind turbine blade. System boundaries include raw material recovery, chemical processing, energy consumption, and disposal alternatives. The impact categories assessed include CO2 emissions, water consumption, energy demand, and waste generation. Data sources include OpenLCA software with secondary data from industry reports and databases. A sensitivity analysis is conducted to assess the variability of results under different energy mix scenarios.

3.2 Data Collection

This study investigates the solvolysis process for wind turbine blade recycling by utilizing a combination of primary and secondary data sources. Primary data was collected through a semi-structured interview with an environmental protection professional from a wind turbine manufacturer, offering critical insights into technological feasibility, operational challenges, and regulatory landscape of blade recycling. The expert's perspective provided a practical understanding of the industry's efforts to minimize environmental impact, enhance energy efficiency, and integrate circular economy principles. Secondary data was obtained from academic literature, industry reports, and regulatory documents, covering key topics such as life cycle assessment (LCA) methodologies, life cycle cost analysis (LCCA), and legislative frameworks governing wind turbine end-of-life management.

4 Results

4.1 Environmental Impact Analysis

Conducting an LCA provides key visualizations, such as model graphs and Sankey diagrams, which offer a comprehensive overview of material and energy flows. The model graph that is generated in the software illustrates the interconnected processes involved in each end-of-life of the blades, allowing them to track inputs, outputs, and environmental burdens systematically (Fig. 1).

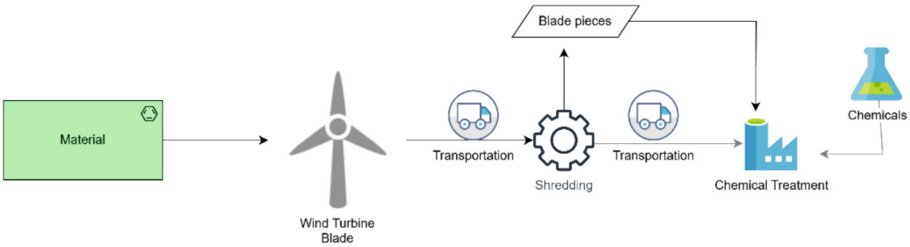

Fig. 1. Value Chain Model graph of Treated Composite Blade

Sankey diagrams, on the other hand, visually represent the magnitude of environmental impacts across different stages, highlighting spots where resource consumption and emissions are most significant. These tools enable a clearer interpretation of complex data, supporting a more transparent comparison between solvolysis and landfill disposal of wind turbine blades (Fig. 2).

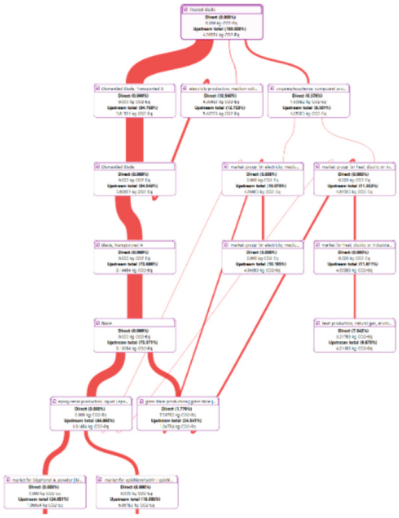

Fig. 2. Sankey diagram from Treated Blade showing the impact of each process/component in the recycling process

The environmental implications of wind turbine blade disposal are a crucial factor in evaluating end-of-life management strategies. Figure 3, compares the environmental implications of two end-of-life management options for wind turbine blades: landfill disposal and solvolysis treatment (treated blade) with EF v3.1 impact assessment method. The evaluation assesses a variety of environmental effect categories, such as water usage, particulate matter generation, material resource depletion, and several toxicity and climate change indicators [16].

The results highlight the environmental burdens across multiple impact categories:

Climate Change and Acidification

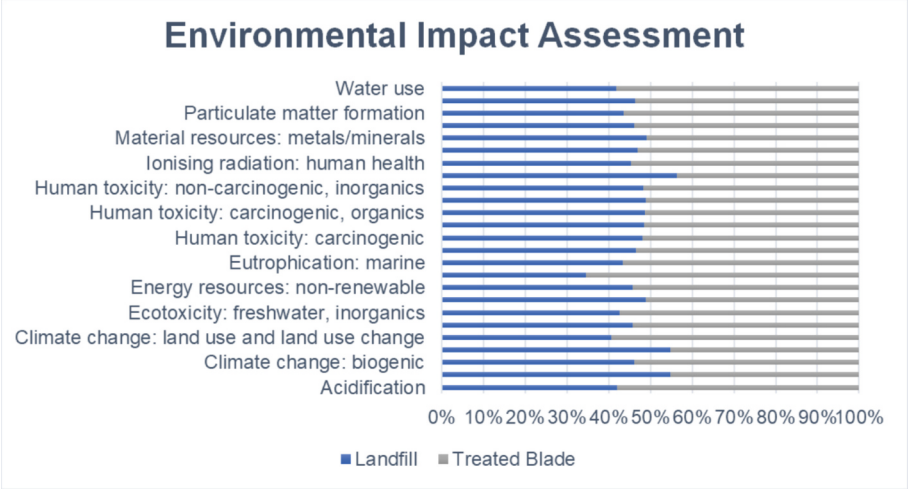

Fig. 3. Environmental Impact Assessment with EF v3.1 as impact assessment method

Solvolysis has more climate change related impacts, including land usage and biogenic contributions, than landfill disposal. The increased energy consumption in solvolysis, along with emissions from chemical inputs, results in a larger carbon footprint. Even though, the overall climate change is less than the landfill's. Moreover, acidification, caused by acidic component emissions, is also increased in solvolysis, indicating the chemical processes involved in breaking down the blade materials.

Water Usage and Energy Resources

Solvolysis treatment uses more water and drains energy resources than landfill disposal. This is mostly owing to the energy-intensive procedures and the usage of enormous amounts of chemicals like isopropanol and organophosphorus compounds in solvolysis. The increased energy demand results in a larger reliance on nonrenewable energy supplies, increasing environmental impact.

Human Health and Toxicity

Solvolysis treatment has a greater influence on human toxicity (both carcinogenic and non-carcinogenic) and ionizing radiation. These effects are caused by the creation and use of chemicals throughout the treatment procedure. In contrast, the landfill process has very low toxicity consequences, because it involves less active processing. However, landfills produce pollutants such as methane and carbon dioxide, which contribute to both local environmental damage and worldwide climate change.

Material Resources and Eutrophication

Material shortage (metals and minerals) is more severe in the solvolysis treatment due to the increased inputs required for chemical processes and the necessity for specialized catalysts. Solvolysis has a stronger eutrophication impact in marine habitats, due to probable chemical emissions and waste byproducts.

Overall, while solvolysis provides a path toward resource recovery and circularity, it also could impose high environmental costs in certain impact categories, particularly

in water use, toxicity, and acidification. These factors must be weighed against the long-term sustainability challenges associated with landfill disposal to determine the most effective strategy for wind turbine blade end-of-life management.

4.2 Economic Feasibility

The economic feasibility of solvolysis versus landfill disposal could be easily evaluated by using the software for the LCA. OpenLCA and SimaPro software tools provide a solid foundation for conducting Life Cycle Cost Analysis (LCCA) by including economic evaluations into Life Cycle Assessment (LCA) either manually or through databases. These tools allow users to attribute monetary values to various phases of a product's life cycle. So, it makes it easier to assess the financial viability of various end-of-life techniques by including cost data directly into environmental models. For instance, OpenLCA, allows for scenario-based cost comparisons, which aid in assessing differences in processing costs, transportation distances, and regulatory fees. This method allows decision-makers to examine not just direct economic costs but also prospective income from material recovery and indirect expenses, such as carbon fees or landfill limitations.

While short-term costs may favor landfill disposal, this option often neglects long-term liabilities and the lost value of wasted materials. On the contrary, solvolysis presents an opportunity to recover valuable materials like glass fiber and epoxy resin, which can generate revenue and help offset operational expenses. This makes it a more economically sound long-term solution. Preliminary theoretical investigations suggest that solvolysis is a more cost-effective option than landfilling. However, further verification of pilot studies is necessary to confirm these findings.

For the early calculation of the total profit per process the following equation is adopted:

$$\textbf{Total Profit} = \textbf{Outputs} - \textbf{Inputs}$$

Total Profit (Landfill) = - License cost for the landfill
Total Profit (Treatment) = Revenue from Recovered Materials - (Cost of Materials + Energy consumption)

While solvolysis offers a compelling economic case for material recovery, its profitability relies on energy price swings and catalyst reuse efficiency which requires future pilot investigations to validate these hypotheses under industrial settings. The high cost of specialized infrastructure, such as industrial autoclaves, prevents broad adoption till nowadays. Furthermore, scaling up the process presents new technological challenges, such as maintaining consistent catalyst performance, reducing operational waste, and optimizing energy use. Addressing these concerns through targeted investments and collaborative initiatives is critical to ensuring that solvolysis delivers economic and environmental benefits in industrial settings.

In conclusion, the preliminary theoretical economic feasibility analysis showed that solvolysis is a more cost-effective choice for wind turbine blade disposal than landfill. While solvolysis needs a significant initial investment due to high input costs, especially

for specialist catalysts and chemicals, its capacity to recover valuable materials such as bisphenol A (BPA), glass fiber (GF), and metals makes up for these costs. In contrast, landfill disposal results in a direct financial loss due to waste processing and disposal expenses, with no possibility of material recovery.

4.3 DPP Enabled Business Ecosystem

Addressing the end-of-life management of wind turbine blades necessitates an integrated recycling ecosystem. This ecosystem involves collaboration among manufacturers, material suppliers, recycling operators, transportation companies, and regulatory bodies. The Digital Product Passport (DPP) could enhance traceability and compliance across the entire value chain network, promoting a digital circular economy and facilitating transparency in material recovery and reuse. Figure 4 illustrates a circular business ecosystem in which different players work together to ensure circularity in wind turbine blade materials. This ecosystem is made up of multiple major entities, each with a critical function aiding material recovery, regulatory compliance, and digital traceability.

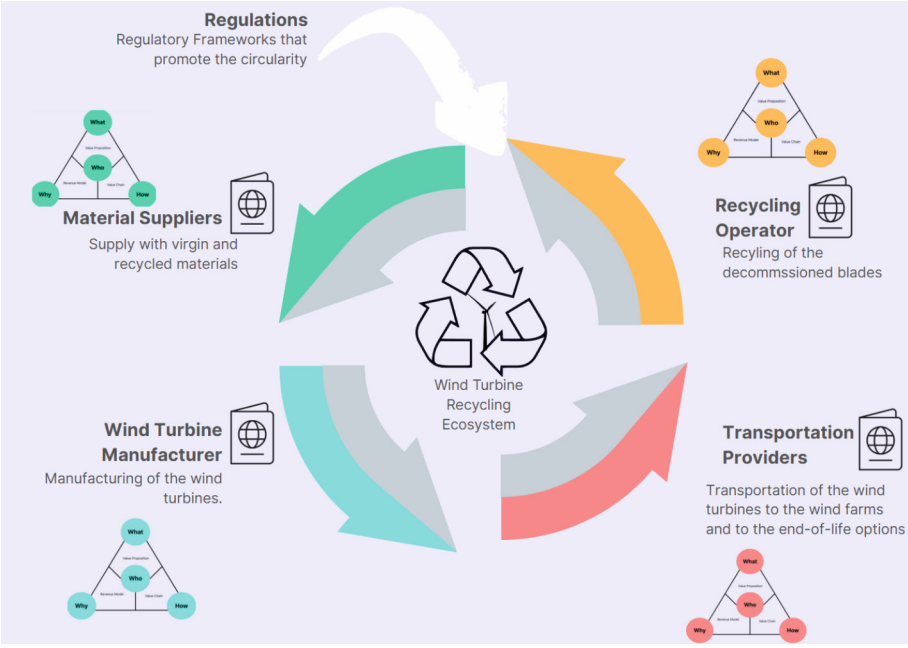

Fig. 4. Business Ecosystem of blade recycling value chain for solvolysis

The wind turbine manufacturer is crucial to the business ecosystem since it manufactures wind turbine blades. The DPP enhances value creation since it enables traceability throughout the product's lifespan, allowing stakeholders to view material composition, maintenance history, and end-of-life treatments. For that reason, the manufacturer should

work with material suppliers, transportation companies, and recycling companies to create an efficient recycling process. Furthermore, the introduction of the recycling service could create new revenue streams from the sale or reuse of materials in the wind turbine.

Material suppliers contribute to the upstream supply chain by providing raw materials for wind turbine blade manufacturing. They could also contribute to the circular economy by reusing materials from decommissioned blades to create novel products, positioning themselves as sustainable suppliers. Furthermore, the integration of DPP allows for effective material tracking, assuring compliance with sustainability norms.

The recycling operator oversees the processing decommissioned blades using modern recycling procedures like solvolysis to recover precious components like fiberglass and epoxy. With DPP's data help, it is possible to optimize the recycling process by offering insights into material qualities and previous changes. Recovered materials could be then returned to material providers or the wind turbine manufacturer, completing the circularity loop.

Transportation companies support the transportation of blades from wind farms to recycling facilities. Given the logistical problems of carrying big wind turbine blades, effective route planning and specialized equipment to dismantle blades are required to save costs and environmental impact. Details provided in the DPP can improve cooperation among producers, recycling operators, and transport providers, facilitating the efficient movement of materials and minimizing the environmental impact of wind turbine blades.

Regulatory frameworks shape the business environment by establishing criteria for sustainable decommissioning and material recovery constraints. Compliance with these requirements is critical to maintaining a legally authorized recycling operation. Regulations also encourage the use of DPP, increasing openness across the supply chain.

The DPP could nurture a business ecosystem that thrives on a circular value exchange. Manufacturers could benefit from using recycled materials, reducing their dependence on virgin resources and lowering production costs. Recycling companies could gain economically by recovering and processing waste materials, turning them into valuable resources. Material suppliers enhance their ecological credentials by providing secondary raw materials for manufacturing. Ensuring the viability of this ecosystem requires regulatory compliance and logistical optimization, which streamline the collection, sorting, and transportation of materials.

The wind turbine blade recycling ecosystem is a collaborative network driven by sustainability, eco-design principles, digitization, and regulatory compliance. DPP as a carrier of LCA and LCCA can promote data exchange efficiency, transparency and material traceability, in a digital industry 4.0/5.0 fashion which strengthens the circular economy human centric collaborative business trust and value generation. This integrated system highlights how industry stakeholders may collaborate to reduce waste, reduce emissions, and improve resource efficiency in the wind energy sector.

4.3.1 Implementation Roadmap of DPP

The effective execution of a Digital Product Passport for wind turbines needs a methodical, strategic 3 phase plan that transitions from theoretical comprehension to a concrete, functional system.

The first phase is the Foundation Phase outlines the strategic framework for the initiative, including defining the blueprint of data points for end-of-life processes like solvolysis and data point requirements for LCA/LCCA. Furthermore, the design of a comprehensive data governance architecture, specifying ownership and security measures. This phase ensures that collaboration between key stakeholders, such as manufacturers, material suppliers, and recycling operators, must be established to enable interoperability and data exchange in strategized industry 4.0 fashion. After the groundwork, phase 2 takes place with a Pilot system development for early validation before a full-scale launch. This prototype enables stakeholders to test functionality and identify issues in real-world settings, which helps to improve the data model, streamline operations, and guarantee user-friendliness. Finally, the Integration phase takes place where the DPP is integrated with enterprise systems like Enterprise Resource Planning (ERP) and Warehouse Management (WHM) via Application Programming Interfaces (APIs). This phase leads to automated exchange of data flow at production ready systems, guaranteeing uniform application of the DPP and emphasizes continues improvements, including the integration of advanced elements such as carbon footprint metrics with IoT or the implementation of blockchain technology for improved security and traceability (Fig. 5).

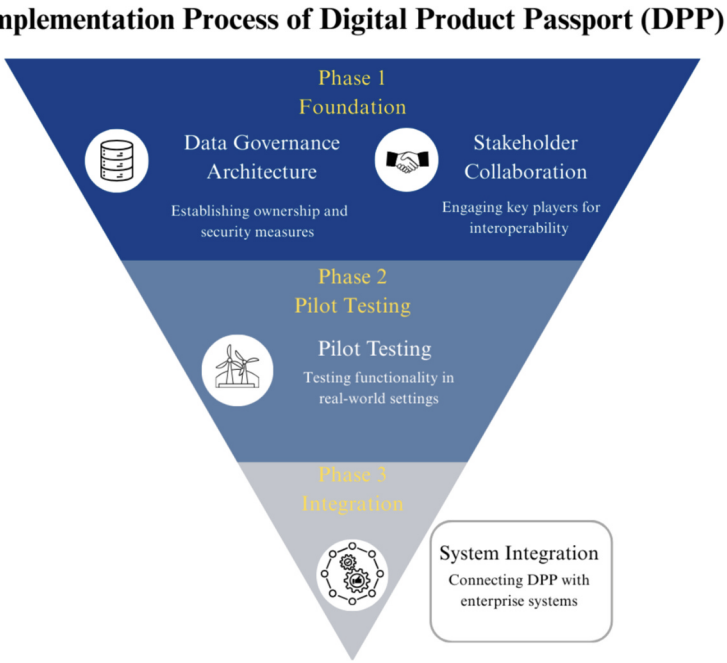

Fig. 5. Implementation Process of Digital Product Passport

5 Conclusions

In conclusion, this study shows that using a chemical recycling method for wind turbine blades could significantly improve sustainability by reducing waste and helping with material recovery. The theoretical LCA findings show that solvolysis has a lower environmental effect than landfilling, especially in terms of decreased emissions and resource saving. Similarly, the LCCA demonstrates that, while the initial investment in recycling infrastructure is significant, the long-term benefits, such as recovered materials and regulatory incentives, could make it a feasible economic option, but further verification using pilot studies are needed. Furthermore, DPP could serve as a digital carrier of LCA and LCCA to increase traceability, optimize end-of-life decision-making, and allow for a more circular approach to blade management.

Transitioning from a linear disposal method to a circular system raises both obstacles and possibilities in terms of business models. While the addition of recycling services necessitates strategic changes to logistics, relationships, and financial planning, it also creates new income streams through material recovery, reuse, and compliance with new environmental standards. Companies who embrace this transformation will gain a competitive advantage in the renewable energy market by proving their dedication to sustainability and innovation.

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